

Mahjong Game Product Requirement Document

Table of Contents

- 1. High-Level Overview
- 2. Core Game Loop & Gameplay Mechanics
- 3. Game Modes & Progression (Tutorial, Quick Play, Full Rotation, Economy, Progression)
- 4. Multi-Ruleset Architecture (Taiwanese, Hong Kong, Riichi)
- 5. UI/UX, Player Assist Features & User Flow Mapping
- 6. AI Behavior & Difficulty Tiers
- 7. Store & Monetization
- 8. Content Strategy & Live Operations
- 9. Go-To-Market (GTM) Strategy
- 10. Appendix A: Comprehensive Ruleset Reference

1. High-Level Overview

This document specifies a highly scalable Mahjong game platform. The default implementation targets the fast-paced, high-scoring Taiwanese 16-Tile ruleset, built on an architecture designed to easily swap modules for Hong Kong 13-Tile and Japanese Riichi variants in future updates.

2. Core Game Loop & Gameplay Mechanics

The core engine manages a state machine tracking a standard deck of 144 tiles (136 suit/honor tiles + 8 Flower/Season tiles). A single round follows this lifecycle:

1. Initialization: Shuffling the wall, rolling dice for the break point, dealing 16 tiles to each player (17 to the dealer).
2. Flower Replacement: Immediate automated identification and replacement of Flower tiles from the back of the wall.
3. Turn Draw: Active player draws a tile. Check for immediate self-drawn win (Zimo) or Quad (Kong).
4. Discard & Interception Window: Active player discards a tile. The game pauses briefly to evaluate interception requests from other players based on standard priority rules: Win (Chong/Ron) > Triplet/Quad (Pong/Kong) > Sequence (Chow).

3. Game Modes & Progression

A. Basic Play Modes

- Onboarding (Tutorial Round): A scripted, three-chapter open-hand tutorial covering fundamental hand composition (5 sets + 1 eye), melding/stealing mechanics, and safe vs.

- dangerous discards.
- Quick Play: A single-round hand with stakes dictated by the player's lobby selection. Ends instantly with score/currency tallies after a win or a wall draw.
- Full Game (Dealer Rotation): A complete session spanning 4 rounds (East, South, West, North winds). Features Taiwanese dealer continuation (Lianzhuang) mechanics that compound stakes for consecutive dealer wins or Tilted draws.

B. The Economy: Virtual Currency & Entry Gates

To establish a compelling long-term engagement loop similar to WSOP Poker, a virtual currency system governs table access. The currency is themed around traditional high-value metrics rather than simple "money" or western "chips".

Primary In-Game Currency: Tael (兩 - Liǎng) / Jade Chips

- Players must wager and possess a minimum balance of Taels to sit at any public table.
- Because Taiwanese Mahjong features wild point variance depending on the winning hand's Tai depth, players must have enough Taels to cover the maximum potential liability of a high-value hand or a base multiplier loss at that specific tier.

C. Lobby Tiers & Stake Levels

Lobbies are segmented by stakes to naturally separate player skill levels and protect the game economy. Lower tiers act as a safety net, while higher tiers demand elite play and calculated risks.

Lobby Name	Entry Requirement	Base Stake per Tai	Max Hand Liability Cap	Target Skill Bracket
Bamboo Courtyard	1,000 Taels	10 Taels	300 Taels	Novice / Free-to-Play Reset Pool
Sparrow Pavillion	10,000 Taels	100 Taels	3,000 Taels	Casual Intermediate
Wind & Cloud Lounge	100,000 Taels	1,000 Taels	30,000 Taels	Advanced / Competitive Regulars
Dragon's Den	1,000,000 Taels	10,000 Taels	No Cap (True Taiwanese Scarcity)	High-Roller / Master Tier

D. Dual-Track Progression & Player Metrics

To incentivize both active play time and skill mastery, progression is split into two distinct tracks:

1. Master Account Level (Experience Points - XP)

- How it accumulates: Gained universally by completing games, winning hands, or declaring complex melds (such as Quads/Kongs). Bonus XP chunks are rewarded through milestone Achievements (e.g., "Win 50 matches", "Complete a hand with an all-triplet composition").
- Player Value: Unlocks aesthetic prestige, customized titles, and access to new lobby environments. It acts as an indicator of absolute lifetime time-spent.

2. Skill Rating System (ELO Matrix)

- How it calculates: A persistent zero-sum matchmaking rating (MMR) that adjusts strictly based on round placements (1st, 2nd, 3rd, 4th). Winning against higher-rated players grants a larger ELO surge, while placing 4th against low-rated opponents severely penalizes the rating.
- Statistical Profile Dashboard: To satisfy detail-oriented competitive players, a comprehensive statistics panel tracks:
 - Win / Loss Ratio: Breakdown of 1st-to-4th finish percentages.
 - Zimo vs. Chong Rate: Measures what percentage of victories were self-drawn compared to catching an opponent's mistake.
 - Deal-In (Feed) Percentage: Tracks how often a player discards the winning tile into an opponent's hand—the core metric for measuring defensive mastery.

4. Multi-Ruleset Architecture

To support future variants, the core board state is strictly separated from the ruleset engine:

Ruleset Variant	Hand Size	Winning Composition	Unique Scoring Metric
Taiwanese (Default)	16 Tiles	5 Melds + 1 Pair (17 Tiles total)	Tai (台) with compounding dealer multipliers
Hong Kong	13 Tiles	4 Melds + 1 Pair (14 Tiles total)	Fan (番) with a minimum point threshold requirement
Japanese Riichi	13 Tiles	4 Melds + 1 Pair (14 Tiles total)	Han & Fu; requires a valid Yaku pattern; uses Dora

5. UI/UX, Player Assist Features & User Flow Mapping

A. Player Assist Features

- Ting (Ready Hand) Indicator: Visually displays when a player is one tile away from winning, listing the required tiles and their remaining quantity in the visible pool.
- Discard Legibility: Selecting a tile in hand highlights identical matches in discard piles and

- public melds. The most recent discard features a distinct glowing border.
- **Automation Toggles:** Settings menu features to auto-arrange hands by suit and automatically declare/draw replacements for Flower tiles.

B. User Flow & Interface Specification

Phase 1: First-Time User Experience (FTUE)

- **Splash & Loading Screen:** Publisher/Studio Logo, Game Title Animation, Universal Progress Loading Bar. Handles asset pre-loading, initial handshake with game server, patch/version checking.
- **Account Creation / Identity Gate:** Anonymous "Guest Play" button, Social Sign-In options (Google, Apple ID, WeChat, Facebook), Terms of Service/Privacy Policy acceptance checkboxes.
- **The Onboarding Choice:** A mandatory modal popup fork for new accounts:
 - Option A: "I'm new to Mahjong" → Directs player straight into the Scripted Tutorial.
 - Option B: "I know how to play" → Skips tutorial, grants initial free-tier onboarding currency, and drops player into the Main Menu.
- **Scripted Tutorial Round:** Dimmed background with bright, forced visual highlights on specific UI elements. Includes dialogue text boxes featuring character guides, step-by-step confirmation prompts, and an omnipresent "Skip Tutorial" button.

Phase 2: Main Menu Hub

- **Main Lobby Dashboard (The Anchor Screen):** Header bar UI featuring Player Avatar, Username, Master Level Badge + Progress Bar, Skill Rating (ELO), Primary Currency Display (Taels/Jade Chips), Premium Currency Display, Settings Icon (Gear), and Mail/Notification Icon. Body layout holds carousel banners for live events/promotions, and primary navigation blocks for game modes.
- **Mode Selection - Quick Play:** Scrollable lobby browser cards representing Stake Tiers (Bamboo Courtyard, Sparrow Pavilion, Wind & Cloud Lounge, Dragon's Den). Displays Tier Name, Entry Fee Requirement, Base Stake per Tai/Fan, Max Hand Liability Cap, Online Player Count Indicator, and Locked/Unlocked state.
- **Mode Selection - Full Round Play:** Structural layout identical to Quick Play, but features distinct labeling indicating a 4-Round Rotation (East to North Wind) and explicit warnings regarding estimated time commitment/quitting penalties.
- **Personalization Dashboard (The Customization Room):** Tabbed interface for Table Cosmetics (swatches for table textures and 3D backdrops), Tile Customization (unlocked/locked skin carousels), and Social Presence (Avatar Frame selector, Title selector, and audio Voice Pack previews).
- **Game Store & Economy Hub:** Tabbed interface for Currency Shop (Premium currency / Tael top-offs), Cosmetic Vault (daily rotating cosmetic assets), and Welfare / Recovery Tab (Alms package redemption button and Ad-Watch video trigger widget).
- **Friends & Social Panel:** Search bar by Player ID, Active Friends list (Online/In-Match status), Pending Friend Invites tab, and "Invite to Lobby" action buttons.
- **Notification & Mail Center:** Tabbed list split into System Announcements and Rewards Mail featuring a universal "Claim All" button.

Phase 3: The Core Game Screen (In-Match UI/UX)

- **Core Table Overlay Elements:** Match State HUD displaying the current prevailing Wind, Round number, Lianzhuang counter, and visual remaining tile count. Opponent Node Widgets (x3) group pods displaying opponent's Avatar, Username, current Tael stack, and an active turn countdown timer bar (3.0s mask / 10s total move limit).
- **The Spatial Table Matrix:** Features the Hidden Hand Area (horizontal row of tile faces for the local player), the Melded Area (dedicated layout space showing public structural melds), and the Center Discard Grid (chronological center grid layout tracking tiles discarded per player).
- **Real-Time Action & Intercept HUD:** Highlight Tracker casts a visual highlight aura over all identical matches when selecting a tile in hand. Interception Pop-up Row contextually animates action buttons into the lower center field (CHOW / PON / KONG / RON / PASS) with a visual ticking countdown bar. The Ting (Ready Hand) Utility Panel displays an expandable assistive slide-out listing required target tiles and their unaccounted-for pool quantities.

Phase 4: Post-Match End Flow

- **Round Scoring Modal (The Tally Sheet):** Comprehensive breakdown of the winning hand's structure. Lists achieved Tai or Fan/Yaku point conditions with corresponding point values. Includes a mathematical calculation ticker demonstrating the payout formula based on table stakes, concluding with a summary of currency shifting across all 4 players' balances.
- **Match End / Rotation Summary:** Final standing podium (1st through 4th place) showing total Taels won/lost during the session, ELO rating adjustments (+/-), and earned Player Account XP animations. Provides buttons for "Play Again" or "Return to Lobby".

6. AI Behavior & Difficulty Tiers

- **Easy (The Randomizer):** Makes basic, uncalculated sets and discards isolated tiles with zero tactical awareness.
- **Medium (The Efficiency Engine):** Calculates hand efficiency, holding connected sets (e.g., 4-5 sequence weights) and prioritizing rapid progression toward Ting.
- **Hard (The Tactician):** Scans opponent discards and open melds. Actively abandons its own hand progression to discard 100% safe tiles if an opponent signals a high-value or ready hand.

7. Store & Monetization

- **Cosmetic Microtransactions:** Custom tile skins (classic jade, modern minimalist, cyberpunk neon), table felt textures, and unique avatar borders or hand-movement animations.
- **Battle Pass / Season Pass:** Tiered progression unlocking limited-edition table assets, background music, and specific voice packs for declaring "Pong!" or "Zimo!".
- **Ticketed Tournament Entries:** Premium tournaments utilizing an entry-fee currency,

offering leaderboard status and premium cosmetic rewards.

8. Content Strategy & Live Operations

- Seasonal Events: Themed events matching regional cultural celebrations (e.g., Lunar New Year variants, Autumn Mid-Autumn cosmetics).
- Dynamic Challenges: Daily and weekly missions incentivizing distinct playstyles (e.g., "Win a hand with a pure suit flush" or "Successfully intercept 3 Pongs in a single match").
- Rule Module Drops: Periodic rollout of secondary rulesets (HK first, Riichi second) alongside regional voice packs and localized aesthetic themes.

9. Go-To-Market (GTM) Strategy

- Target Demographics: Casual board game enthusiasts looking for quick mobile/web play sessions, alongside core Mahjong players seeking highly accurate rule representations.
- Beta Phase Deployment: Launching a free web-based prototype focusing entirely on the Taiwanese Quick Play loop to stress-test the rules engine and gather UI feedback.
- Community & Influencer Alignment: Partnering with digital board game streamers and community groups to host low-stakes virtual tournaments, leveraging the spectator-friendly nature of high-tier Mahjong gameplay.

10. Appendix A: Comprehensive Ruleset Reference

To preserve clean architectural separation between system design and mathematical scoring logic, this section houses the technical requirements for the three planned engine variants.

A. Taiwanese Mahjong (Default Engine)

- Tile Deck Composition: 144 Tiles total (36 Characters, 36 Bamboos, 36 Dots, 16 Winds, 12 Dragons, 8 Flowers/Seasons).
- Hand Configuration: 16 hidden tiles active. A complete winning hand requires exactly 17 tiles (5 Melds $\times$ 3 tiles + 1 Eye $\times$ 2 tiles).
- The Draw Cycle: Flower tiles are mechanically "unplayable" in the hidden hand. Drawing a Flower tile triggers an immediate automation state that places it in the public pool and draws a replacement tile from the back of the wall before checking turn states.
- The Multiplier Base (Tail - 台): Hand values are strictly additive based on patterns, but the dealer status implements exponential scaling.
- Lianzhuang (Dealer Continuation): If the East player retains their dealer status, a base compounding formula applies to the final round value: $\text{Base Hand Tai} + (2k + 1)$, where k represents the consecutive counter of the dealer's streak.
- Muck/Draw Rules: Unlike other variants where a dry wall triggers an empty hand, if a Taiwanese round concludes in a draw, the active dealer's hand state is verified. If the dealer was Ting (Ready), they retain the deal and increment the counter. If they were not Ting, the deal passes to the right but the streak reset occurs.

B. Hong Kong Mahjong (First Expansion Module)

- Tile Deck Composition: Options for 144 Tiles (including Flowers for casual rules) or 136 Tiles (strict competitive standard removing Flowers entirely). Default engine module will support an admin toggle for "Casual/Competitive Flower Allocation".
- Hand Configuration: 13 hidden tiles active. A complete winning hand requires exactly 14 tiles (4 Melds $\times$ 3 tiles + 1 Eye $\times$ 2 tiles).
- Scoring Architecture (Fan - 番): Values scale exponentially up to a strict limit (4 Fan $\rightarrow$ 8 Fan $\rightarrow$ 16 Fan or modern limits up to 10 Fan).
- Minimum Threshold Requirement (Fan Gate): Lobbies can implement a strict "Minimum Fan Requirement" (commonly 1 Fan or 3 Fan). If a player completes a valid physical hand configuration but their calculated score pattern yields a value lower than the gate threshold, the engine blocks the "Win" selection entirely. Clicking a tile that results in a sub-threshold win is classified as a foul hand (Chombo).

C. Japanese Riichi Mahjong (Advanced Strategy Module)

- Tile Deck Composition: 136 Tiles strictly (No Flower/Season tiles used). Features 4 "Akadora" (Red Five replacement tiles: one Red 5-Pin, one Red 5-Sou, two Red 5-Wan) which act as passive point boosters if held in a winning hand.
- Hand Configuration: 13 hidden tiles active. A complete winning hand requires exactly 14 tiles (4 Melds + 1 Eye).
- The Yaku Validity Constraint: A hand cannot declare victory purely because it is geometrically complete ($4 \times 3 + 2$). It must possess at least one valid Yaku (a recognized scoring pattern, such as All Simples, Riichi, or an Honor Triplet).
- Special Engine Mechanics:
 - The Riichi Declaration: If a player has a closed hand and reaches Ting, they can wager 1,000 points, slide a tile sideways into the discard pool, and lock their hand into an automated "Draw-or-Discard Only" state. This unlocks the ability to hit the "Riichi" Yaku value.
 - Furiten (Sacred Discards): If a player is Ting, but any of their winning target tiles are currently sitting in their own historical discard pile, they enter a Furiten State. While in Furiten, the game engine completely disables the "Ron" (Win off an opponent's discard) button. The player can only win via "Tsumo" (Self-draw).
 - Dora Tracking: One tile in the dead wall is flipped face-up at the start of the match. The numerical tile immediately following it sequentially becomes the "Dora" tile, adding passive hand value if carried to a valid victory.